Fiscal Year:	FY 2024	**Task Last Updated:**	FY 02/24/2025
PI Name:	Olson, Sandra Ph.D.		
Project Title:	Fundamental Research on International Standard of Fire Safety in Space - Subteam 1: Study of Flammability of Fabric Materials		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	Sandra.Olson@nasa.gov	**Fax:**	FY 216 977-7065
PI Organization Type:	NASA CENTER	**Phone:**	216-433-2859
Organization Name:	NASA Glenn Research Center		
PI Address 1:	LTX, Combustion Physics and Reacting Systems Branch		
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City:	Cleveland	**State:**	OH
Zip Code:	44135	**Congressional District:**	9
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2012 Japanese Space Agency (JAXA) AO for Fundamental Research on an International Standard of Fire Safety in Space
Start Date:	07/01/2014	**End Date:**	12/30/2025
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Urban, David	**Contact Phone:**	216-433-2835
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 12/30/2025 per PI (Ed., 2/7/25) NOTE: End date changed to 3/31/2024 per PI (Ed., 10/29/21) NOTE: End date changed to 7/31/2022 per PI (Ed., 10/5/20) NOTE: Changed end date to 3/31/2021 per PI (Ed., 6/3/19)		
Key Personnel Changes/Previous PI:	Dr. Sandra Olson is U.S. Co-Investigator on Japan Aerospace Exploration Agency (JAXA)-sponsored project, “Flammability Limits At Reduced-g Experiment (FLARE)." JAXA Principal Investigator (PI) is Prof. Osamu Fujita, Hokkaido University.		
COI Name (Institution):			
Grant/Contract No.:	Internal Project		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	The objective of the project is to develop a methodology to correlate material flammability limits in normal gravity and microgravity, which allows quantitative estimation of material flammability limit in microgravity based on the flammability data obtained on the ground. The project involves an international team including Japan Aerospace Exploration Agency (JAXA), NASA, ESA (European Space Agency), and universities in Japan, USA, and France. Dr. Olson is a U.S. Co-Investigator for the JAXA sponsored experiments to be conducted aboard the Japanese Experiment Module, Kibo. To establish global standards for fire safety in space, we seek to develop a fundamental understanding of how NASA's material flammability test, NASA-STD-6001.A Test 1, relates to the actual flammability of materials in micro and partial gravity. The investigation strategy is to perform extensive research via ground-based experiments, including 1g and parabolic flight tests, and via theoretical formulations. Flight experiments on orbit in International Space Station (ISS)/KIBO will be performed to verify the correlation. The flight experiments on orbit are expected in 2018 or later. By the end of the project, a new fire safety standard test method for screening spacecraft materials will be proposed that addresses the shortcomings of existing standard test methods such as NASA-STD-6001B.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Studying materials flammability in space allows us to accurately control the flow field and thus elucidate the importance of a critical Damkohler number (flow time /reaction time) on flame extinction. The anticipated improved methodology should reduce time and cost for the spacecraft material screening. Investigation and results have Earth benefits for terrestrial fire safety.
Task Progress:	ISS experiments are underway. Experiment 1-2 filter paper samples were completed in March 2024. The PMMA samples for Experiment 1-1 were conducted starting in March 2024. Since another experiment was scheduled to use the vacuum resources needed for the FLARE experiments in June 2024, the PMMA samples were finished in August 2024 and the rod samples for Experiment 2-1 were started in August 2024. Unfortunately, the sample holder became jammed after the first rod sample, so the decision was made to move onto other samples while this issue was considered. Experiment 1-4 Nomex samples were run starting in September 2024. There was another conflict with another experiment, so operations were suspended in December and January. In February, a gas bottle leak was discovered, and troubleshooting is underway. Experiment 1-3 cheesecloth samples will start in March 2025. The annual international FLARE Workshop was held at JAXA in Tsukuba, Japan in September 2024 where results and plans were discussed by the international science team.
Bibliography Type:	Description: (Last Updated: 01/12/2026)
Abstracts for Journals and Proceedings	Li C, James S, Ferkul PV, Olson SL, Johnston MC. "Extinction of solid diffusion flame in microgravity: Details of quenching and blowoff processes. " 2024 Flammability Limits At Reduced-g Experiment (FLARE) Workshop, Tsukuba, Japan, September 4-6, 2024. Sponsored by Japan Aerospace Exploration Agency (JAXA). 2024 Flammability Limits At Reduced-g Experiment (FLARE) Workshop, Tsukuba, Japan, September 4-6, 2024. Sponsored by Japan Aerospace Exploration Agency (JAXA). , Sep-2024
Articles in Peer-reviewed Journals	Li C, James S, Ferkul PV, Olson SL, Johnston MC. "Extinction of solid diffusion flame in microgravity: Details of quenching and blowoff processes." Proceedings of the Combustion Institute. 2024;40(1-4):105373. https://doi.org/10.1016/j.proci.2024.105373 , Jul-2024